MILITARY SPECIFICATION

TRANSMITTERS, PRESSURE, HELICOPTER, VOLTAGE RATIO TYPE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for voltage ratio type helicopter pressure transmitters.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-C-7742	Screw Threads, Standard; Optimum Selected Series. General Specification for
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-I-58098	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification for

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage

FSC 6620

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MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards; Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MS33586	Metals, Definition of Dissimilar
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Internal Mounting

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing. The type of testing shall be as specified in the detail specification.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitter except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of a corrosion-resistant type or be suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions encountered in storage or normal service.

3.3.4 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion. Dissimilar metals are defined in MS33586.

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3.3.5 Toxic and corrosive fumes. The transmitter shall be of such material and construction that when burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.6 Protective treatment. Materials used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, shall be protected against such deterioration in a manner that will in no way prevent compliance with performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

3.4.1 Design. The transmitter shall be designed to sense the differential of two pressures and to convert differential pressure measurement into an electrical signal for use of indicators conforming to MIL-I-58098. The transmitter shall not require use of an amplifier to provide this signal to the indicator. The transmitter shall be of simple design and shall be easy to maintain. Bearings, gears, or parts that are subject to frictional wear or brinelling shall not be used. Elastic constraints shall be used for guidance and control of all moving parts.

3.4.1.1 Adjustments. The transmitter shall not require periodic adjustments, such as for manually set bridge-balance potentiometer However, a means shall be provided for adjusting and correcting indications over a minimum total of 10 percent of full-scale which shall be simple to manipulate and of such design that adjustments can be made with a minimum of special tools.

3.4.1.2 Operating position The transmitter shall operate as specified herein in either vertical or horizontal positions.

3.4.1.3 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover within a range of 105 to 130 precent of full-scale deflection

3.4.1.4 Lubrication. The transmitter shall be designed so that no periodic lubrication will be required during service.

3.4.2 Construction. The transmitter shall be so constructed that no parts will work loose in service. It shall be built to withstand strains, jars, vibrations, and other conditions incident to handling, shipment, storage, installation, and service.

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3.4.2.1 Noncorrosive sealing. The electrical system of the transmitter shall be isolated from both pressure compartments and atmosphere by a hermetic seal. The pressure sensing element and any part in contact with the pressure medium shall be constructed of materials that will resist corrosive actions of alcohol, aircraft fuels and oils, and hydraulic fluid.

3.5 Performance. The transmitter shall operate as specified herein after subjection to the following conditions:

a. Temperatures - operation at temperatures ranging from -54° to $+165^{\circ}C$ and storage at temperatures ranging from -62° to $+232^{\circ}C$

b. Vibration - vibration of 20g's from 5 to 2,000 Hz

c. Temperature-altitude - pressure equivalent to a 35,000 foot altitude at $-55^{\circ}C$

d. Humidity - relative humidity of 95 percent

e Salt fog - exposure to salt sea atmosphere for 48 hours

f. Fungus - fungus growth encountered in tropical climates

3.5.1 Tolerances The transmitter shall operate within tolerances specified in the detail specification.

3.5.2 Damping. The time required for change from full-scale pointer deflection (251°) to 35° shall not exceed 2 ± 1 second under the conditions specified in section 4

3.5.3 Vent boss and connector strength. The vent boss and connector shall withstand 25 foot-pounds bending moment applied to the vent boss and connector without evidence of calibration shifting out of tolerance, distortion of base plate, or damage to welds and soldering.

3.5.4 Overpressure. The transmitter shall withstand a 100 percent overpressure for a period of 10 minutes without exceed scale error tolerance specified in the detail specification.

3.5.5 Case leakage. For hermetically sealed transmitters, the case leakage shall not exceed 0.1 micron cubic foot per hour.

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3.5.6 Dielectric strength. The transmitter shall withstand application of 200V rms at a commercial frequency for a period of 5 minutes.

3.5.7 Power supply variations. The transmitter shall meet the power supply variations of MIL-STD-704 within tolerances specified in the detail specification.

3.5.8 Radio noise interference. The transmitter shall meet the radiated and conducted radio noise interference requirements of MIL-STD-461.

3.5.9 Endurance. The transmitter shall withstand 60,000 applications of cycling from zero indication to 80 ± 5 percent and back to zero at a rate of 30 ± 10 cpm.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Item identification and part number requirements of MIL-STD-100 shall govern manufacturer's part numbers and changes thereto.

3.7 Case. The case shall conform to Figure 1.

3.7.1 Filling medium. When case is hermetically sealed, the filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. Other concentrations of filling medium shall not be used except with prior approval of the procuring activity. Absolute pressure of filling medium in the case shall be 12 to 14.7 psi.

3.8 Electrical connector. An MS33678R10SL-3P electrical connector shall be provided. Connector shall have polarizing key at the top and shall conform to MIL-C-5015.

3.9 Electrical characteristics.

3.9.1 Transducer. Transducer shall consist of a magnetic armature connected to the prime mover and centrally positioned to the field of two concentrically wound induction coils. Coil wiring shall be in accordance with Figure 2.

3.9.2 The transmitter shall meet the following requirements with a normal excitation of 26V 400 Hz single phase applied.

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3.9.2.1 No load. With terminal C open, current used by the transmitter shall not exceed 160 ma.

3.9.2.2 Power consumption. The transmitter power consumption shall not exceed 0.9 watt.

3.9.2.3 Secondary voltage. The maximum voltage measured between terminals A and C and between B and C shall not exceed 15V.

3.9.2.4 Open circuit tap voltage. The open circuit tap voltage and phase angle at 0, 50% and 100% of full-scale pressure shall be measured using the circuit described in Figure 4. The tap voltage and phase angle between indicator terminals A and C resulting from these measurements shall be calculated and shall be within the following limits:

Percent Deflection	Tap Voltage Ratio	Phase Angle
0	.5896 ± .0100	0 ± 2.5
50	.4953 ± .0100	0 ± 2.5
100	.4066 ± .0100	0 ± 2.5

3.9.2.5 Impedance. The Thevenin impedance of the transmitter at 0, 50% and 100% of full-scale pressure shall be measured using the maxwell bridge method, (see Figure 5). The impedance resulting from these measurements shall be calculated and shall be within the following limits. The bridge input voltage shall be 26.0 ± 0.05V, 400.0 ± 0.1 Hz Sine wave with 0.5% maximum distortion.

Percent Deflection	Resistance (ohms)	Inductance (millihenrys)
0	19.75 ± 1.98	28.87 ± 2.89
50	21.68 ± 2.17	30.50 ± 3.05
100	20.68 ± 2.30	29.17 ± 2.92

3.9.3 External wiring. External wiring shall be in accordance with Figure 3.

3.10 Power supply. The transmitter shall operate from a 26 ± 2v 400 ± 10-Hz single-phase power source having wave form, crest factor, and harmonics in accordance with MIL-STD-704.

3.11 Prime mover. The prime mover shall be a pressure-sensing element of which the motion shall be directly detected without mechanical amplification.

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3.12 Vent connection. Vent pressure connections shall be provided with a threaded boss as shown on Figure 1 and shall be connected to the opposing pressure element. The connection shall be marked VENT in durable white letters 1/8 inch high.

3.13 Finishes and protective coatings.

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

3.13.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.13.1.2 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14 Weight. Weight of the transmitter shall not exceed 1.5 pounds.

3.15 Screw threads. Unless otherwise specified, threads of all machine screws shall conform to MIL-S-7742.

3.16 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.16.1 Nameplate. A nameplate shall be permanently attached to the rear of case. Where applicable, information marked therein shall conform to MIL-STD-130. Nameplate shall not be glued or cemented to the case nor shall metal decals be used.

3.17 Workmanship. The transmitter shall be constructed and finished in a workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as in consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability or performance of the indicator, they shall be held or limited accordingly.

3.17.2 Cleaning. The transmitter shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, or other foreign material removed during and after final assembly. Burrs, sharp edges, and resin flash that might crumble shall also be removed.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of transmitters shall be classified as acceptable tests and as specified in the detail specification.

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Wherever pressure or temperature existing at the time of the test is not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) or at room temperature (approximately 25°C) When tests are made with atmospheric pressure or room temperature differing materially from above values, proper allowance shall be made for the difference from specified condition.

4.3.2 Tapping Unless otherwise specified, the transmitter shall be lightly tapped or vibrated before each reading.

4.3.3 Test voltage. Unless otherwise specified, the transmitter shall be tested with the voltage and frequency specified in 3.10. The voltage and frequency shall be maintained within ±1 percent of the nominal value.

4.3.4 Test standard. The transmitter shall be tested with an indicating test standard consisting of a standard indicator unit calibrated to a maximum error of ±¼ percent and conforming to the requirements of MIL-I-58098.

4.4 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests
b. Sampling tests

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4.4.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.5:

a. Examination of product
b. Dielectric strength
c. Scale and friction error
d. Position error
e. Sealing

4.4.2 Sampling tests.

4.4.2.1 Sampling plan A. One transmitter shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.5·

a. Individual tests
b. Vibration
c. Low temperature
d. High temperature

4.4.2.2 Sampling plan B. Unless otherwise specified, 3 transmitters shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.5:

a. Sampling plan A tests
b. Electrical characteristics
c. Seasoning
d. Endurance
e. Temperature-altitude
f. Low temperature exposure
g. High temperature exposure
h. Humidity
i. Salt fog
j. Fungus
k. Radio noise interference
l. Power supply variations
m. Vent boss and connector strength
n. Overpressure
o. Damping

4.4.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until extent and cause of failure have been determined and appropriately corrected.

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4.4.2.4 Individual tests may continue. Individual tests may be continued pending investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that items meet all requirements of the specification.

4.4.3 Defects in items already accepted. Investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test methods.

4.5.1 Examination of product. The transmitter shall be examined to determine compliance with requirements specified herein with respect to materials, markings, dimensions, and workmanship.

4.5.2 Dielectric strength. A potential of 200V at a commercial frequency shall be applied between any pin of the electrical receptacle and transmitter case for a minimum of 5 minutes. There shall be no insulation breakdown.

4.5.3 Scale and friction error. The transmitter shall be tested for scale error at ambient temperature at test points specified in the detail specification. Transmitter shall be lightly tapped or vibrated before each reading to remove normal friction. The difference between the readings before and after transmitter is tapped or vibrated shall constitute friction error. Scale and friction errors shall not exceed tolerances specified in the detail specification. This test shall be conducted both up-scale and down-scale at the specified test points

4.5.4 Position error. The transmitter shall be energized to obtain a scale reading of approximately one-half scale. Maximum change in indication produced by rotating the transmitter from normal operating position to a position 90° both clockwise and counter-clockwise (longitudinal axis vertical) shall not exceed position error tolerance specified in the detail specification. .

4.5.5 Sealing.

4.5.5.1 Nonhermetically sealed case. The pressure and vent connections of nonhermetically sealed case shall be closed with plugs and the transmitter immersed in a de-aerated liquid, such as water, within an enclosure. Absolute pressure of the air above the liquid

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shall be reduced to 2½ inches Hg. Any bubbles coming from within the case shall be considered leakage and shall be cause for rejection. Bubbles that are a result of entrapped air in exterior parts of the case will not be considered leakage. A wetting agent shall be used in the liquid to facilitate removal of surface bubbles.

4.5.5.2 Hermetically sealed case. The hermetically sealed case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. Detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.5.6 Vibration. The transmitter shall be subjected to a vibration test in accordance with MIL-STD-810, method 514.1, equipment category (b), curve L of Figure 514.1-2, procedure I, part 1, at an indication of approximately midscale. Pointer oscillation and variation shall not exceed tolerances specified in the detail specification. At conclusion of this test, the transmitter shall be subjected to the scale error test and shall meet tolerances specified in the detail specification. No damage to the indicator shall result from this test.

4.5.7 Low temperature. The transmitter shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502, procedure I, at $-54^{0} \pm 2^{0}C$ for a period of 4 hours. At end of this period and while still at specified temperature, the indicator shall be subjected to scale and friction error test. Errors shall not exceed tolerances specified in the detail specification.

4.5.8 High temperature. The transmitter shall be subjected to a high temperature test in accordance wtih MIL-STD-810, method 501, procedure I, at $165^{0} \pm 2^{0}C$ for a period of 6 hours. During this test, transmitter shall be electrically connected to a standard indicator with power applied At end of the 6 hour period and while still at specified temperature, the transmitter shall be subjected to scale and friction error test Tolerances shall not exceed those specified in the detail specification.

4.5.9 Electrical characteristics. The transmitter shall be tested to determine compliance with 3.9.2.1 through 3.9.2.5.

4.5.10 Seasoning. The transmitter shall be operated for 10,000 cycles at a rate of 30 ± 10 cpm from zero indication to 80 ± 5 percent of full-scale and back to zero indication Within 1 hour after conclusion of this test, the transmitter shall be subjected to scale error test. Errors shall not exceed tolerances specified in the detail specification. The transmitter shall then be examined to determine that no undue wear has occurred.

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4.5.11 Endurance. The transmitter shall be subjected to an endurance test in accordance with 4.5.10 except it shall be operated for 60,000 cycles in lieu of 10,000 cycles.

4.5.12 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 504, procedure I (except temperature shall be -55°C) with chamber pressure equivalent to an altitude of 35,000 feet. Rate of pressure change shall not exceed an average of 3 psi per second while under these conditions. At the end of a 1-hour period, a scale and friction error test shall be conducted. Errors shall not exceed tolerances specified in the detail specification. This test may be conducted concurrently with the low temperature exposure test.

4.5.13 Low temperature exposure. The low temperature exposure test shall be conducted in accordance with MIL-STD-810, method 502, procedure I. Transmitter shall be exposed to a temperature of -62° plus or minus 2°C for a minimum of 48 hours after which it shall be operated throughout its range while at the specified temperature. Temperature shall then be raised to -54° ± 2°C and the transmitter maintained at this temperature for a minimum of 24 hours. At end of 24 hour period, the transmitter output shall be within low-temperature accuracy tolerances specified in the detail specification. Transmitter shall be returned to room temperature for a minimum period of 4 hours and then subjected to the scale error test. Scale errors shall not exceed tolerances specified in the detail specification for room temperature.

4.5.14 High temperature exposure. The transmitter shall be placed in a temperature chamber and subjected to 232° ± 10°C for a period of 80 hours. It shall then be returned to room temperature for a minimum of 4 hours after which it shall be subjected to scale error test. Error shall not exceed room temperature tolerances specified in the detail specification.

4.5.15 Humidity. Nonhermetically sealed transmitters shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, procedure I. External connections to the transmitter shall be made to simulate installed conditions. Immediately after completion of this test, the transmitter shall be subjected to dielectric strength, and scale and friction error tests. Errors shall not exceed tolerances specified in the detail specification.

4.5.16 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I, for a period of 50 hours. The transmitter shall be examined and there

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shall be no evidence of external corrosion or deterioration that would affect subsequent operation. External surface of the transmitter may be washed with distilled water and air-dried after which it shall be subjected to the dielectric strength test. Measured resistance shall be a minimum of 20 megohms.

4.5.17 Fungus. Nonhermetically sealed transmitters shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I, after completion of the qualification or preproduction tests (as applicable). The instrument shall previously have been subjected to salt fog test. The transmitter shall be disassembled so that internal mechanism will be exposed to the fungus spores At end of the test period, the transmitter shall be examined to determine that no fungus growth has occurred.

4.5.18 Radio noise interference. The transmitter shall be subjected to a radio noise interference test in accordance with MIL-STD-461 except that radiated noise shall be within the range of 0.15 to 150 megacycles.

4.5.19 Power supply variations. The transmitter shall meet scale error and friction tolerances specified in the detail specification when subjected to power supply variations as specified in MIL-STD-704.

4.5.20 Vent boss and connector strength The vent boss and connector mounting shall be sufficiently rigid to withstand a force of 25 foot-pounds bending moment applied in any direction perpendicular to longitudinal axis of the transmitter without evidence of calibration shifting-out-of tolerance, distortion of base plate, or damage to vent boss and connector welds or soldering. This test shall be conducted with the transmitter secured by engine mount connection or adjacent hex nut only. The transmitter shall then be subjected to sealing test at ambient temperature. Scale error shall not exceed tolerances for ambient temperature specified in the detail specification.

4.5.21 Overpressure. The transmitter shall be subjected to 100 percent overpressure for a period of 10 minutes. At least 1 hour after conclusion of this test, the transmitter shall be subjected to scale error test using increasing pressure only. Error shall not exceed tolerances specified in the detail specification.

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4.5.22 Damping. A pressure sufficient to produce full-scale deflection shall be applied to the transmitter and released by a quick-release valve. Time required for change from 25^{0} (full-scale deflection) to 35^{0} shall be 2 ± 1 seconds. The transmitter shall be tested in a test rig conforming to Figure 6 with transmitter in an upright position (vent at the top and pressure connection at the bottom.) Lubricating oil used for this test shall conform to MIL-L-7808 and shall have a viscosity of approximately 2,500 centistokes at -54^{0}C. Test to be performed at room temperature.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be as specified in the contract or order (see 6.2).

5.1.1 Reinspection date. Reinspection date markings shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use The transmitters covered by this specification are intended for use with variable-reluctance-type indicators conforming to MIL-I-58098 in providing remote indication of aircraft pressure functions.

6.2 Ordering data. Ordering data shall be as specified in the detail specification.

6 2 1 Contract data requirements. Contract data requirements shall be as specified in the detail specification.

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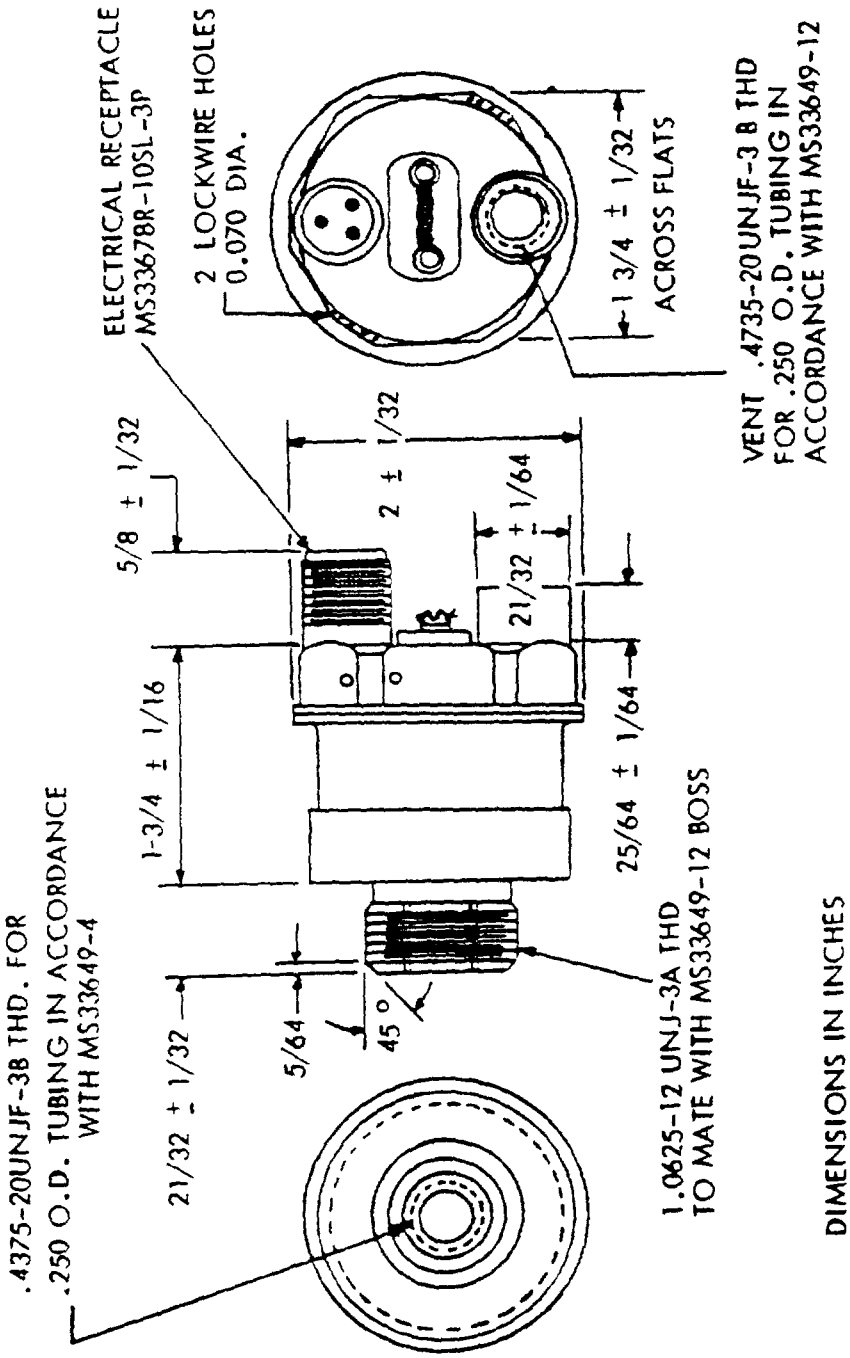
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PREPARING ACTIVITY:

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Project No. 6620-A003

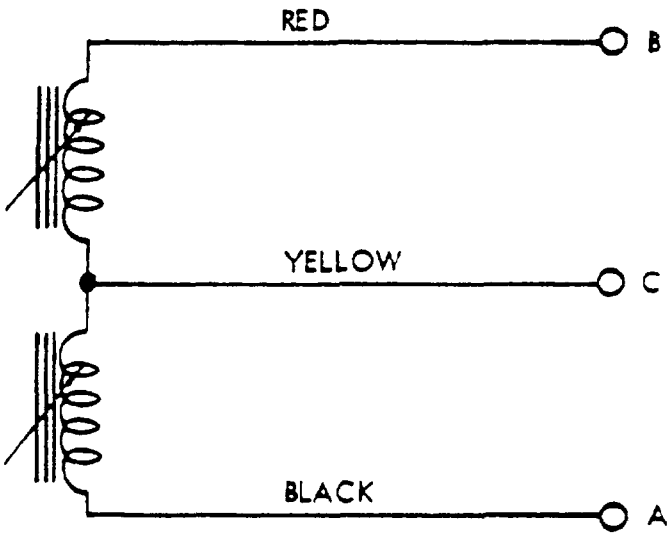
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DIMENSIONS IN INCHES

FIGURE 1. TRANSMITTER CASE

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A & B - POWER 26V 400~ 1ϕ
A & C - TAP VOLTAGE

FIGURE 2. TRANSMITTER INTERNAL WIRING

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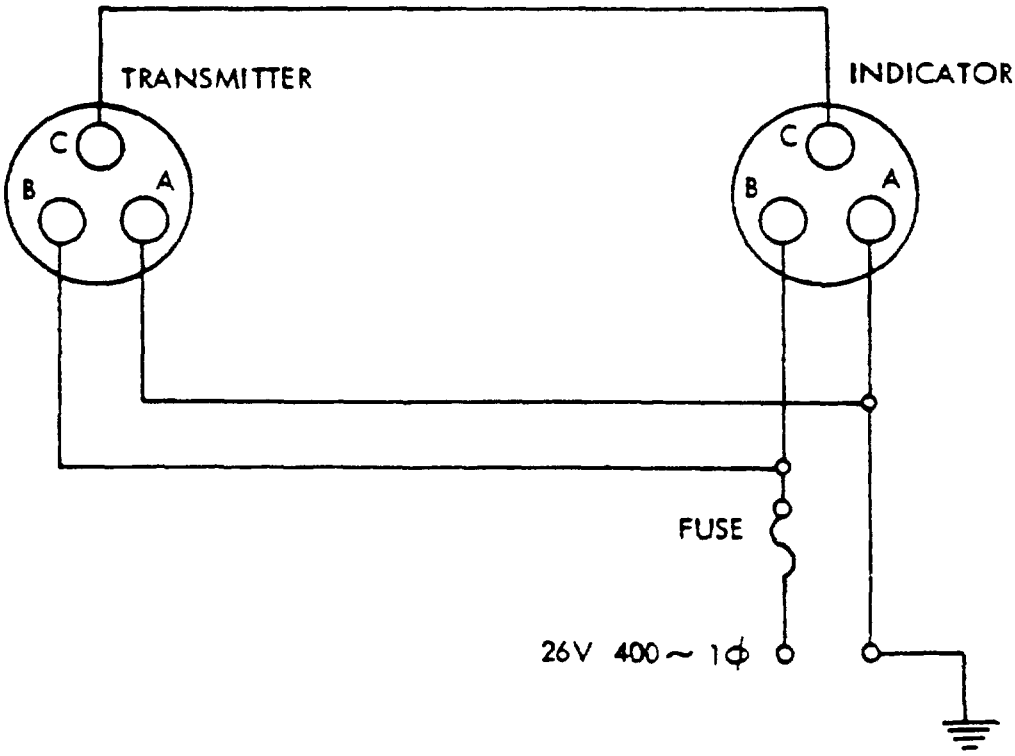
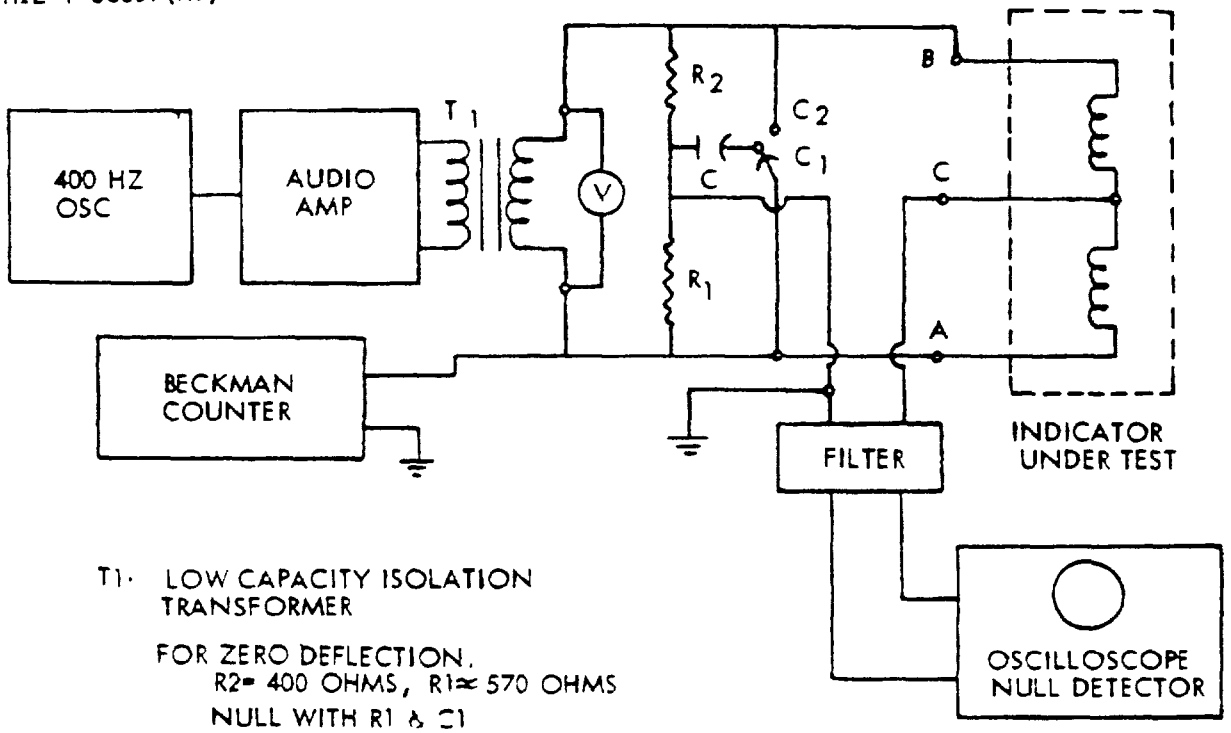


FIGURE 3. EXTERNAL WIRING

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T1. LOW CAPACITY ISOLATION TRANSFORMER

FOR ZERO DEFLECTION,
R2 = 400 OHMS, R1 ≈ 570 OHMS
NULL WITH R1 & C1

FOR FULL SCALE DEFLECTION
R1 = 100 OHMS, R2 ≈ 570 OHMS
NULL WITH R2 & C2

BRIDGE INPUT = 26.0V ± 0.05V, 400.0 ± 0.1 HZ SINE WAVE 0.5% MAXIMUM DISTORTION

$$X = \frac{R1\,(R1 + R2 + R1 \times R2^2 \times \omega^2 \times C2^2)}{(R1 + R2)^2 + (R1 \times R2 \times \omega)^2\,(C1^2 + C2^2)}$$

$$Y = \frac{R1 \times R2 \times \omega \times (R2 \times C2 - R1 \times C1)}{(R1 + R2)^2 + (R1 \times R2 \times \omega)^2\,(C1^2 + C2^2)}$$

$$E = \sqrt{X^2 + Y^2}$$

$$\phi L = \frac{180}{\pi}\ \text{ARC TAN}\ \frac{Y}{X}$$

FIGURE 4. TAP VOLTAGE MEASURING BRIDGE

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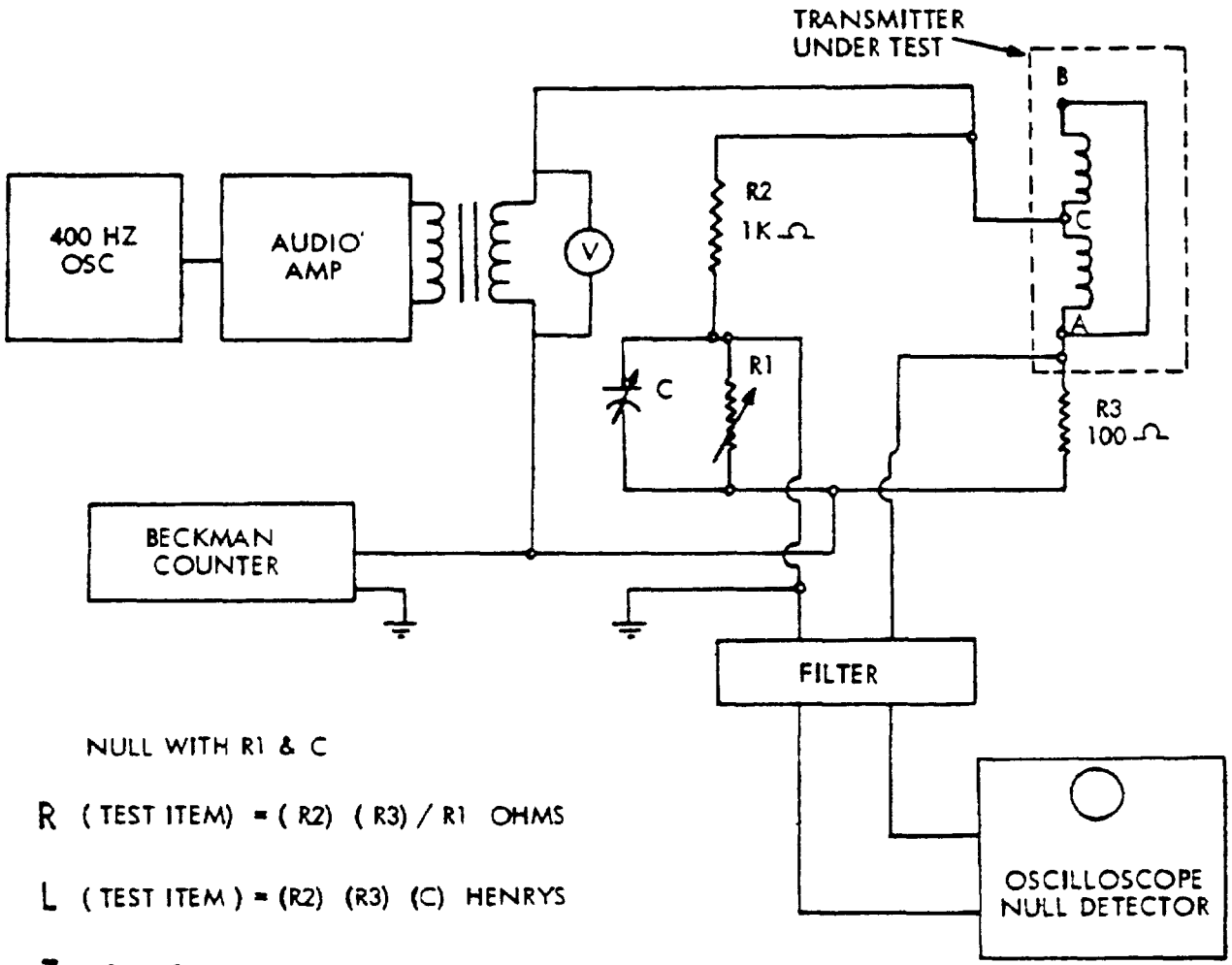
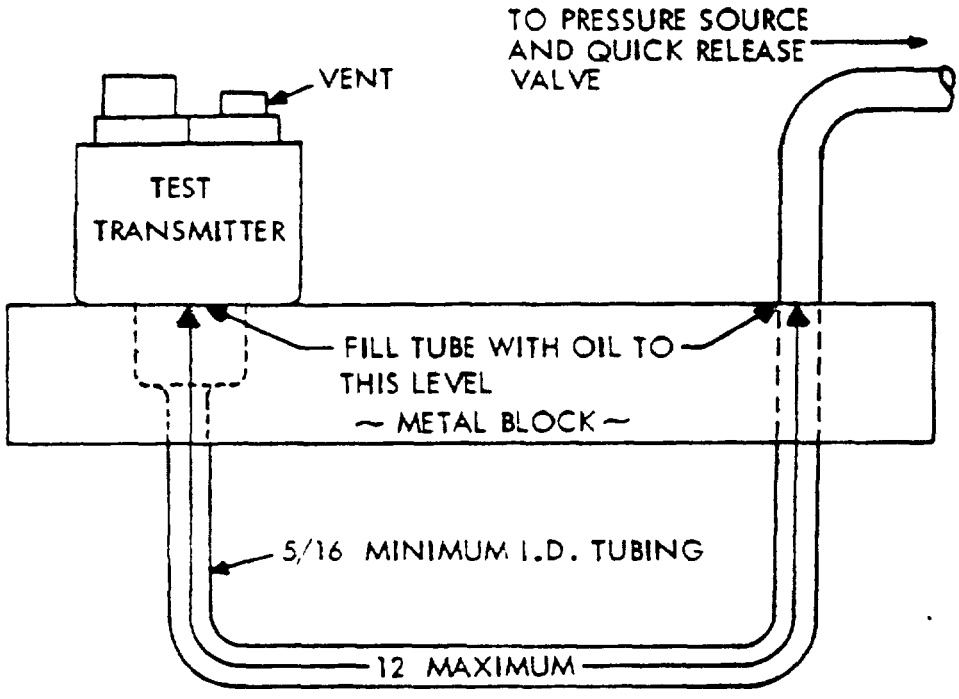


FIGURE 5. THEVENIN IMPEDANCE MEASURING BRIDGE

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DIMENSIONS IN INCHES

FIGURE 6. TEST RIG FOR DAMPING TEST

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